BERTHOUMIEU Emile-Marie

Current position : PhD student at CERFACS
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After graduating with a Franco-Canadian double degree following a year-and-a-half-long research project. I started a PhD in hydrogen combustion modeling in 2023. Currently applying for my first postdoc position with a view to continuing my career in combustion research.

BACKGROUND

Megep *2023-Today*
Toulouse, France
Second year in an PhD program in Energy science.

ETS *2019-2022*
Montréal, Canada
Third year in an engineering program in a master's degree specialized in aerospace.

ENSEIRB-MATMECA *2017-2019*
Bordeaux INP, France
Two years in an engineering program specializing in mathematics, mechanics and numerical simulation

SCIENTIFIC PROJECTS

PhD: Theoretical analysis of diffusion flames and development of a Large Eddy Simulation model for turbulent hydrogen diffusion flames *2023-Today*
CERFACS, Toulouse, France
Development of a new model dedicated to non-premixed flames for large-scale simulations using a physical and mathematical approach

Simulation and modeling of the impact of structure orientation on thermal conduction during laser-matter interaction *2020-2022*
ETS/ArianeGroup Research Chair, Montréal, Canada
Development of an original multi-scale physical model and its numerical implementation via difference methods

Simulation of acoustic wave scattering in a highly dispersive medium *2018-2019*
ENSEIRB-MATMECA, Bordeaux, France
Implementation of a numerical solution of a three-dimensional heat equation in cylinders via different numerical approaches such as a finite volume method, finite differences or the use of ABAQUS.

RESEARCH OVERVIEW

Hydrogen become increasingly present in our daily lives. However, introducing hydrogen in aeronautic engines leads to several challenges. One difficulty is that, because of the high reactivity of hydrogen, many hydrogen air flames burn in non-premixed modes. This leads to an higher needs of modeling on those flames. My actual research focuses on a better understanding of these flames. Based on these behaviors, create a new numerical model valid for any pressure or temperature.

SKILLS

Solid mechanics: Resistance of materials, Laws of behavior, Dynamics of mechanical structures, vibration and fatigue theory damage and failure
Fluid mechanics: Combustion, Dimensional analysis , Navier-Stokes equations
Numerical analysis: Finite differences, Finite volumes, Finite elements
Numerical simulation: AVBP, CANTERA, ABAQUS, ANSYS WorkBench Fluent
Computer Science: Python, Fortran 90 , C++, LaTeX
Linguistics: French (mother tongue), English, Japanese (basics)

APPENDICES

Publication list
BERTHOUMIEU, Emile-Marie, POINSOT, Thierry, BERTIER, Nicolas. Spatial under-resolution and thickening models for LES of turbulent diffusion flames. Combustion And Flame. Accepted and Under processing

Conferences
European Combustion Meeting, 2025. Edinburgh, England.
INCA Symposium, 2024. Marseille, France